Data, Distributions, and Correlation

The Basics

Experimental unit — The object on which information is collected. This information is organized into observed variable values.

Univariate data — Information measured on one variable

Multivariate data — Information measured on multiple variables

In this course, we focus on understanding the **relationships that exist in multivariate data**, particularly how variables interact and influence one another in applied contexts.

Example: Financial Data (`stock_returns.csv`)

Consider a dataset consisting of daily stock returns for multiple companies:

- NVIDIA (NVDA)
- Apple (AAPL)
- Amazon (AMZN)
- AMD (AMD)

Each row represents a trading day, and each column represents the return (%) of a stock.

Sample of the Data

Day	NVDA	AAPL	AMZN	AMD
1	1.2	0.8	0.6	1.5
2	-0.5	0.3	-0.2	-0.8
3	2.1	1.0	0.9	2.5
4	0.7	0.4	0.5	1.1
5	-1.3	-0.6	-0.9	-1.7
		…		
40	-0.6	-0.3	-0.4	-1.0

Data Description

The data consists of 40 observations (trading days) and 4 variables (stock returns).
This dataset is multivariate because multiple variables are measured simultaneously for each observation.

Notes

1. Each row represents a trading day (experimental unit).
2. Each column represents a different measured variable (stock return).

Independence Assumption

For most methods in this course, experimental units are assumed to be independent.
Here, independence means that returns on one trading day do not directly influence another day.
Although financial data may show temporal patterns, we assume independence for introductory analysis.

Research Questions

- Do these stocks move together?
- Which stock is most volatile?
- Are there strong correlations among these variables?
- How does covariance affect portfolio risk?

Example: Financial Asset Structure

Consider a dataset where each asset is described using multiple characteristics.

Sample Data (`asset_structure.csv`)

Interpretation

Each variable captures a different aspect of asset behavior.
Researchers are often interested in grouping assets that have **similar characteristics**, even if their values differ in magnitude.

Research Questions

Asset	Return	Volatility	Volume	Momentum	Beta
NVDA	1.2	2.5	1.8	0.9	1.4
AAPL	0.8	1.6	1.5	0.7	1.1
AMZN	0.6	1.8	1.6	0.8	1.2
AMD	1.5	2.8	2.0	1.0	1.5
TSLA	2.0	3.2	2.3	1.3	1.8

- Which assets behave similarly?
- Can assets be grouped based on volatility and return?
- Do high-beta assets cluster together?

Example: Market Decision Data

Consider a dataset describing trading decisions or financial outcomes.

Sample Data (`market_decisions.csv`)

ID	Return	Market	Volatility	Liquidity	Condition	Success
1	1.2	0.9	2.5	1.8	1	1
2	-0.5	-0.3	2.0	1.5	0	0
3	0.8	0.6	1.7	1.6	1	1
4	-1.0	-0.8	2.8	1.4	0	0
5	1.5	1.2	3.0	2.1	1	1

Interpretation

- Each row represents a trading decision (experimental unit).
- Each column represents a variable influencing the outcome.

The response variable **Success** indicates whether the decision was profitable.

Key Questions

- Which variables influence trading success?
- Does volatility affect outcomes?

- Can success be predicted using multiple variables?

There is a large amount of information contained in such datasets.
How can all of this information be summarized, and is any of it meaningful?
What types of inferences can be made from this data set?
These questions motivate the use of multivariate statistical methods, which allow us to analyze multiple variables simultaneously and extract meaningful insights.

Types of Variables

Continuous variables — Numerical values that can occur anywhere within an interval; there is no fixed number of possible values.

Discrete variables — A fixed set of possible values (numerical or nonnumerical).

Example: Financial Data

Day is a discrete variable. Stock returns (NVDA, AAPL, AMZN, AMD) can be treated as continuous variables.

Example: Market Decision Data

Success and market condition variables are discrete variables. Return and volatility can be treated as continuous variables.
Many multivariate methods were developed for continuous variables under a multivariate normal distribution assumption (more on this later).

Some of the Topics for this Semester

Methods to summarize data

- Graphics — We will go beyond simple 2D scatter plots and use techniques such as parallel coordinate plots to visualize multivariate data.
- Principal components analysis and factor analysis — These methods reduce the dimension (number of variables) of a data set by forming new variables that are (ideally) interpretable.

- Cluster analysis — Methods to group previously ungrouped items.

Methods to predict classifications (where classifications are previously known)

- Discriminant analysis
- Nearest neighbor analysis
- Logistic and multinomial regression
- Classification trees

Classical methods

- Hotelling's T^2 provides a multivariate extension of a t-test.
- Multivariate analysis of variance (MANOVA) provides a multivariate extension of analysis of variance (ANOVA).

Data Matrices and Vectors

Notation

- p = number of numerical variables of interest.
- N = number of experimental units on which the variables are being measured.
- x_{rj} = value of the jth variable on the rth experimental unit, for $r = 1, \ldots, N$ and $j = 1, \ldots, p$.
- X = data matrix: the x_{rj} values are arranged so that x_{rj} is the (r, j) entry of X.
- Rows represent experimental units and columns represent variables.
- X has dimension $N \times p$.
- $\mathbf{x}_r$ = vector of data for the rth experimental unit.
- $\mathbf{x}_r = \begin{bmatrix} x_{r1} & x_{r2} & \cdots & x_{rp} \end{bmatrix}'$.

Example: Financial Data

Consider only the variables NVDA, AAPL, AMZN, and AMD (daily returns).
These variables are already expressed in comparable percentage units, so no adjustment (such as scaling) is required.

Sample Data

Day	NVDA	AAPL	AMZN	AMD
1	1.2	0.8	0.6	1.5
2	-0.5	0.3	-0.2	-0.8
3	2.1	1.0	0.9	2.5
⋮	⋮	⋮	⋮	⋮
40	-0.6	-0.3	-0.4	-1.0

The Day variable is included to help identify the experimental units.

Notation for this Data

- $p = 4$
- $N = 40$
- $x_{11} = 1.2, ; x_{12} = 0.8, ; x_{13} = 0.6, ; x_{14} = 1.5$
- $x_{21} = -0.5, ; x_{22} = 0.3, ; \ldots$

The Multivariate Normal Distribution

The normal distribution for a random variable can be generalized to a multivariate normal distribution for a vector of random variables (random vector). Some of the theory for multivariate analysis methods relies on the assumption that a random vector has a multivariate normal distribution.

Univariate Normal Distribution

Let x be a random variable with a univariate normal distribution with mean $E(x) = \mu$ and variance $Var(x) = E[(x - \mu)^2] = \sigma^2$. This is represented symbolically as $x \sim N(\mu, \sigma^2)$.
The probability density function of x is:

$$f(x \mid \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \qquad -\infty < x < \infty$$

Example: Univariate normal distribution plot

Suppose $x \sim N(45, 4^2)$. Below is the code used to plot $f(x|\mu, \sigma)$:

```
> curve(expr = dnorm(x = x, mean = 45, sd = 4), xlim =
> c(30, 60), col = "red", xlab = "x", ylab = "f(x)", main =
> "Univariate normal distribution")
> dnorm(x = c(30, 45, 60), mean = 45, sd = 4)
> abline(h = 0)
>
```

Let

$$\mathbf{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_p \end{bmatrix}$$

has a multivariate normal distribution.
To introduce the multivariate normal distribution function, we need to examine a few other items first:

- Let $E(x_i) = \mu_i$
- The relationships between the x_i's can be measured by the covariances and/or the correlations

Covariance of x_i and x_j

$$\text{Cov}(x_i, x_j) = E\big[(x_i - \mu_i)(x_j - \mu_j)\big] = \sigma_{ij}$$

Variance of x_i

$$\text{Var}(x_i) = E\big[(x_i - \mu_i)^2\big] = \sigma_{ii}$$

Correlation coefficient of x_i and x_j

$$\rho_{ij} = \frac{\sigma_{ij}}{\sqrt{\sigma_{ii}\sigma_{jj}}} = \frac{\text{Cov}(x_i, x_j)}{\sqrt{\text{Var}(x_i)\,\text{Var}(x_j)}}$$

Remember that $\sigma_{ij} = \sigma_{ji}$ and $-1 \leq \rho_{ij} \leq 1$. Another commonly used notation for ρ_{ij} is Corr(x_i, x_j).

Mean vector, covariance matrix, and correlation matrix

The means, covariances, and correlations can be put into a mean vector, covariance matrix, and a correlation matrix:

$$\boldsymbol{\mu} = \mathsf{E}(\mathbf{x}) = \begin{bmatrix} \mathsf{E}(x_1) \\ \vdots \\ \mathsf{E}(x_p) \end{bmatrix} = \begin{bmatrix} \mu_1 \\ \vdots \\ \mu_p \end{bmatrix},$$

$$\Sigma = \mathsf{Cov}(\mathbf{x}) = \mathsf{E}\big[(\mathbf{x} - \boldsymbol{\mu})(\mathbf{x} - \boldsymbol{\mu})'\big] = \begin{bmatrix} \sigma_{11} & \sigma_{12} & \cdots & \sigma_{1p} \\ \sigma_{21} & \sigma_{22} & \cdots & \sigma_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{p1} & \sigma_{p2} & \cdots & \sigma_{pp} \end{bmatrix}, \text{ and}$$

$$P = \mathsf{Corr}(\mathbf{x}) = \begin{bmatrix} 1 & \rho_{12} & \cdots & \rho_{1p} \\ \rho_{21} & 1 & \cdots & \rho_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ \rho_{p1} & \rho_{p2} & \cdots & 1 \end{bmatrix}.$$

Variance Expansion

$$\begin{aligned} \mathsf{Var}(x_i) &= E\big[(x_i - \mu_i)^2\big] \\ &= E\big(x_i^2 - 2x_i\mu_i + \mu_i^2\big) \\ &= E(x_i^2) - 2\mu_i E(x_i) + \mu_i^2 \\ &= E(x_i^2) - 2\mu_i^2 + \mu_i^2 \\ &= E(x_i^2) - \mu_i^2. \end{aligned}$$

Another Way to Express the Covariance Matrix

$$\begin{aligned} \Sigma &= \mathsf{Cov}(\mathbf{x}) \\ &= E\big[(\mathbf{x} - \boldsymbol{\mu})(\mathbf{x} - \boldsymbol{\mu})'\big] \\ &= E\big(\mathbf{x}\mathbf{x}' - \mathbf{x}\boldsymbol{\mu}' - \boldsymbol{\mu}\mathbf{x}' + \boldsymbol{\mu}\boldsymbol{\mu}'\big) \\ &= E\big(\mathbf{x}\mathbf{x}'\big) - E\big(\mathbf{x}\boldsymbol{\mu}'\big) - E\big(\boldsymbol{\mu}\mathbf{x}'\big) + E\big(\boldsymbol{\mu}\boldsymbol{\mu}'\big) \\ &= E\big(\mathbf{x}\mathbf{x}'\big) - \boldsymbol{\mu}\boldsymbol{\mu}'. \end{aligned}$$

Notice that if the mean is **0**, then

$$\Sigma = E\big(\mathbf{x}\mathbf{x}'\big).$$

- Remember that Σ and P are symmetric and knowing Σ is equivalent to knowing P.

Example: Bivariate normal distribution

Suppose

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

has a multivariate normal distribution with

$$\boldsymbol{\mu} = \begin{bmatrix} 12 \\ 18 \end{bmatrix} \quad \text{and} \quad \Sigma = \begin{bmatrix} 1.5 & 0.4 \\ 0.4 & 1.1 \end{bmatrix}.$$

We could write this as $\mathbf{x} \sim N_2(\boldsymbol{\mu}, \Sigma)$.

Correlation matrix

$$P = \begin{bmatrix} 1 & 0.31 \\ 0.31 & 1 \end{bmatrix}$$

For example,

$$\rho_{12} = \frac{0.4}{\sqrt{1.5 \cdot 1.1}} \approx 0.31.$$

Multivariate normal density function

$$f(\mathbf{x} \mid \boldsymbol{\mu}, \Sigma) = \frac{1}{(2\pi)^{p/2}\,|\Sigma|^{1/2}} \exp\big(-\tfrac{1}{2}(\mathbf{x} - \boldsymbol{\mu})'\Sigma^{-1}(\mathbf{x} - \boldsymbol{\mu})\big)\,.$$

R code

```
> library(rgl)
> library(mvtnorm)

> mu <- c(12, 18)

> sigma <- matrix(data = c(1.5, 0.4, 0.4, 1.1), nrow = 2,
> ncol = 2, byrow = TRUE)
```

```
> P <- cov2cor(V = sigma)

> P
>
```

3D Plot and Contour Plot

There is not a `curve()` function in R that will produce a 3D rotatable plot if simply given `dmvnorm()`. Instead, we create a set of possible x_1 and x_2 values which are used to evaluate `dmvnorm()` to obtain $f(x)$. I use these resulting values with a 3D rotatable plotting function and also a contour plotting function. Below is the code and output:

```
> # Calculate f(x) for a large number of possible values for x1 and x2

> x1 <- seq(from = 10, to = 18, by = 0.1)
> x2 <- seq(from = 16, to = 22, by = 0.1)

> all.x <- expand.grid(x1, x2)  # Create all possible combinations

> head(all.x)  # Notice that x1 is changing faster than x2
> Var1 Var2
> 1 10.0   16
> 2 10.1   16
> 3 10.2   16
> 4 10.3   16
> 5 10.4   16
> 6 10.5   16

> eval.fx <- dmvnorm(x = all.x, mean = mu, sigma = sigma)

> # Arrange values in the following form:
>
> #
>
> # x2
>
> # 16.0 16.1 16.2 ...
>
> # x1 10.0 f(x) f(x) f(x)
```

```
>
> # 10.1 f(x) f(x) f(x)
>
> # 10.2 f(x) f(x) f(x)
>
> # ...

> fx <- matrix(data = eval.fx, nrow = length(x1), ncol =
> length(x2), byrow = FALSE)
>


> # Check if f(x) values are arranged correctly
>
> all.x[1:2,]
> Var1 Var2
> 1 10.0   16
> 2 10.1   16

> dmvnorm(x = all.x[1:2,], mean = mu, sigma = sigma)

> fx[1:2,1]  # Part of column 1

> dmvnorm(x = c(10.0, 16), mean = mu, sigma = sigma)

> fx[1,1:2]  # Part of row 1
>


> # 3D plot
>
> persp3d(x = x1, y = x2, z = fx, col = "green",
> xlab = "x1", ylab = "x2", zlab = "f(x)")
>


> # Contour plot - purposely made x and y-axes the same
>
> # length so that one can judge variability
>
> par(pty = "s")
```

```
> contour(x = x1, y = x2, z = fx, main = "Multivariate normal",
> xlab = expression(x[1]), ylab = expression(x[2]),
> xlim = c(9, 19), ylim = c(15, 23))

> # panel.first = grid() does not work quite right with contour()
>
> # so used abline() instead

> abline(h = 16:22, lty = "dotted", col = "lightgray")

> abline(v = seq(from = 10, to = 18, by = 2),
> lty = "dotted", col = "lightgray")
>
```

Interpretation of Contour and Surface Plots

To rotate the first plot, hold your left mouse button down while you move the mouse. To zoom in on the first plot, hold your right mouse button down while you move the mouse.

Examine the following:

1. The surface is centered at $\boldsymbol{\mu}$.
2. The surface is wider in the x_2 direction than in x_1. This is because there is more variability.
3. Notice the shape of the surface. The contour plot is an ellipse and the surface plot looks like a 3D normal curve.
4. The volume underneath the surface is 1.

Note that the 3D plot is not in the graphics window. To get the plot into Word, I used the Windows' Snipping Tool.

Suppose

$$\boldsymbol{\mu} = \begin{bmatrix} 12 \\ 18 \end{bmatrix} \quad \text{and} \quad \Sigma = \begin{bmatrix} 1.5 & 0.4 \\ 0.4 & 2.2 \end{bmatrix}.$$

The only change from the previous example is that σ_{22} has increased. Note that

$$P = \begin{bmatrix} 1 & 0.22 \\ 0.22 & 1 \end{bmatrix}.$$

Below are 3D surface and contour plots drawn on the same scale for the previous and new normal distributions.

Multivariate Summary Statistics

Let $x_1, x_2, \ldots, x_N$ be a random sample from a multivariate distribution (not necessarily normal) with a mean vector $\boldsymbol{\mu}$, covariance matrix Σ, and correlation matrix P. This can be represented symbolically as $x_{i\,i=1}^{N} \sim (\boldsymbol{\mu}, \Sigma)$.
Note on notation: If it is understood that a random sample of size N is taken, then we could shorten this to $\mathbf{x} \sim (\boldsymbol{\mu}, \Sigma)$.

Estimates of $\boldsymbol{\mu}$, Σ, and P based on the random sample are:

$$\hat{\boldsymbol{\mu}} = \frac{1}{N}\sum_{r=1}^{N} \mathbf{x}_r = \frac{1}{N}(\mathbf{x}_1 + \mathbf{x}_2 + \cdots + \mathbf{x}_N),$$

$$\hat{\Sigma} = \frac{1}{N-1}\sum_{r=1}^{N} (\mathbf{x}_r - \hat{\boldsymbol{\mu}})(\mathbf{x}_r - \hat{\boldsymbol{\mu}})'.$$

Notes

$$\hat{\boldsymbol{\mu}} = \begin{bmatrix} \frac{1}{N}\sum_{r=1}^{N} x_{r1} \\ \frac{1}{N}\sum_{r=1}^{N} x_{r2} \\ \vdots \\ \frac{1}{N}\sum_{r=1}^{N} x_{rp} \end{bmatrix} = \begin{bmatrix} \hat{\mu}_1 \\ \hat{\mu}_2 \\ \vdots \\ \hat{\mu}_p \end{bmatrix}$$

$$\hat{\Sigma} = \begin{bmatrix} \hat{\sigma}_{11} & \hat{\sigma}_{12} & \cdots & \hat{\sigma}_{1p} \\ \hat{\sigma}_{21} & \hat{\sigma}_{22} & \cdots & \hat{\sigma}_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ \hat{\sigma}_{p1} & \hat{\sigma}_{p2} & \cdots & \hat{\sigma}_{pp} \end{bmatrix}$$

where

$$\hat{\sigma}_{ii} = \text{Var}(x_i) = \frac{1}{N-1}\sum_{r=1}^{N} (x_{ri} - \hat{\mu}_i)^2,$$

$$\hat{\sigma}_{ij} = \text{Cov}(x_i, x_j) = \frac{1}{N-1}\sum_{r=1}^{N} (x_{ri} - \hat{\mu}_i)(x_{rj} - \hat{\mu}_j).$$

Equivalently, you can express $\hat{\Sigma}$ as follows. Suppose $\tilde{X}$ is the $N \times p$ matrix of mean-adjusted values, i.e., its (r, j) entry is $x_{rj} - \hat{\mu}_j$. Then

$$\hat{\Sigma} = \frac{1}{N-1}\tilde{X}'\tilde{X}.$$

One can see the two expressions for the estimated covariance matrix are equivalent by examining a very small example (e.g., let $p = 2$ and $N = 3$).

$$
\begin{aligned}
r_{ij} &= \frac{\hat{\sigma}_{ij}}{\sqrt{\hat{\sigma}_{ii}\hat{\sigma}_{jj}}} \\
&= \frac{\sum_{r=1}^{N}(x_{ri} - \hat{\mu}_i)(x_{rj} - \hat{\mu}_j)}{\sqrt{\sum_{r=1}^{N}(x_{ri} - \hat{\mu}_i)^2}\sqrt{\sum_{r=1}^{N}(x_{rj} - \hat{\mu}_j)^2}}.
\end{aligned}
$$

$$
R = \begin{bmatrix} 1 & r_{12} & \cdots & r_{1p} \\ r_{21} & 1 & \cdots & r_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ r_{p1} & r_{p2} & \cdots & 1 \end{bmatrix}.
$$

Example: Stock Returns Data (`stock_returns.csv`)

Let x_1 denote NVDA returns, x_2 denote AAPL returns, x_3 denote AMZN returns, and x_4 denote AMD returns. Each variable represents daily percentage returns for a given stock, and observations correspond to trading days.
Let

$$
\mathbf{x}_r = \begin{bmatrix} x_{r1} \\ x_{r2} \\ x_{r3} \\ x_{r4} \end{bmatrix}
$$

be the vector of returns on the rth trading day.
Suppose we have a random sample of size $N = 40$ from $(\boldsymbol{\mu}, \Sigma)$. Thus we observe $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_{40}$.
Using these observations produces the estimates of $\boldsymbol{\mu}$, Σ, and P:

$$
\hat{\boldsymbol{\mu}} = \begin{bmatrix} \hat{\mu}_1 \\ \hat{\mu}_2 \\ \hat{\mu}_3 \\ \hat{\mu}_4 \end{bmatrix} = \begin{bmatrix} 0.85 \\ 0.52 \\ 0.60 \\ 1.10 \end{bmatrix}
$$

$$
\hat{\Sigma} = \begin{bmatrix} 1.95 & 0.72 & 0.68 & 0.95 \\ 0.72 & 1.40 & 0.75 & 0.82 \\ 0.68 & 0.75 & 1.60 & 0.88 \\ 0.95 & 0.82 & 0.88 & 2.10 \end{bmatrix}
$$

$$R = \begin{bmatrix} 1 & 0.43 & 0.38 & 0.47 \\ 0.43 & 1 & 0.50 & 0.48 \\ 0.38 & 0.50 & 1 & 0.44 \\ 0.47 & 0.48 & 0.44 & 1 \end{bmatrix}$$

R code (to compute $\hat{\mu}$, $\hat{\Sigma}$, and R)

```
> stock <- read.csv("stock_returns.csv")

> head(stock)
> Day  NVDA  AAPL  AMZN  AMD
> 1   1  1.20  0.80  0.60  1.50
> 2   2 -0.50  0.30 -0.20 -0.80
> 3   3  2.10  1.00  0.90  2.50
> 4   4  0.70  0.40  0.50  1.10
> 5   5 -1.30 -0.60 -0.90 -1.70
> 6   6  0.90  0.50  0.70  1.30

> X <- as.matrix(stock[, 2:5])

> mu.hat <- colMeans(X)
> mu.hat
> NVDA    AAPL    AMZN     AMD
> 0.8500  0.5200  0.6000  1.1000

> sigma.hat <- cov(X)
> sigma.hat
> NVDA  AAPL  AMZN   AMD
> NVDA   1.950 0.720 0.680 0.950
> AAPL   0.720 1.400 0.750 0.820
> AMZN   0.680 0.750 1.600 0.880
> AMD    0.950 0.820 0.880 2.100

> R <- cor(X)
> R
> NVDA  AAPL  AMZN   AMD
> NVDA   1.000 0.430 0.380 0.470
> AAPL   0.430 1.000 0.500 0.480
> AMZN   0.380 0.500 1.000 0.440
```

```
> AMD    0.470 0.480 0.440 1.000
>
```

Alternative Computation

```
> cov2cor(sigma.hat)

> apply(X, 2, mean)

> X.tilde <- X - matrix(mu.hat, nrow = nrow(X),
                    ncol = ncol(X), byrow = TRUE)

> t(X.tilde) %*% X.tilde / (nrow(X) - 1)
>
```

Simulation Example

```
> library(mvtnorm)

> mu <- c(12, 18)

> sigma <- matrix(data = c(1.5, 0.4, 0.4, 1.1),
              nrow = 2, byrow = TRUE)
> set.seed(1001)

> N <- 20

> x <- rmvnorm(n = N, mean = mu, sigma = sigma)

> mu.hat <- colMeans(x)
> mu.hat

> sigma.hat <- cov(x)
> sigma.hat

> R <- cor(x)
> R
>
```

Compare the estimates to the true values. Are the estimates reasonable? What would happen if a larger sample size was taken?

Sample Size Comparison

```
> N <- 200
>
```

```
> N <- 2000
>
```

Notice how the estimates improve as the sample size increases.

Questions

- Why do the points fall within higher density contour regions?
- How does increasing sample size affect the estimates?
- What happens to the correlation estimates as N increases?

Multivariate normal contour plot with a sample

Notice that the points are within the larger value contours. Why should this happen?
What if $N = 200$ instead? I used the same code as before except changed the value of N:

```
> mu.hat
> [1] 12.1034 17.9821

> sigma.hat
> [,1]     [,2]
> [1,] 1.623450 0.472118
> [2,] 0.472118 1.298762
>
```

What if $N = 2000$ instead?

I used the same code as before except changed the value of N:

```
> mu.hat
> [1] 12.0218 18.0045

> sigma.hat
> [,1]     [,2]
> [1,] 1.512338 0.401225
> [2,] 0.401225 1.102884
>
```

I was a little surprised to see r_{12} get farther from the true value. I also tried $N = 20000$ and obtained $r_{12} = 0.36$.

Questions

- Why do the points fall within higher density contour regions?
- How does increasing sample size affect the estimates?
- Why might correlation estimates fluctuate even for large N?

Checking Multivariate Normality

Some of the statistical methods that we will examine this semester have an underlying multivariate normal distribution assumption for the data.

Questions

1. Suppose a statistical method has a multivariate normal distribution assumption and it is not satisfied. What are the potential consequences?
2. How could you approximately verify multivariate normality for:
    - data collected on two variables?
    - data collected on three variables?
    - data collected on more than three variables?

An *ad hoc* way to check a multivariate normality assumption is to look for data falling into an approximate ellipse shape within each possible pairwise scatter plot (use side-by-side scatter plots — see the graphics section). This does *not* ensure multivariate normality, but it provides some credibility to the assumption.

Example: Bivariate normal distribution plots

For the data simulation example

$$\mathbf{x} \sim N_2\left(\begin{bmatrix}12\\18\end{bmatrix}, \begin{bmatrix}1.5 & 0.4\\0.4 & 1.1\end{bmatrix}\right),$$

below are 3D kernel density estimates of the density of **x**.

Questions

- For $N = 20$ and $N = 200$, rotate the 3D plots. Do the point clouds look approximately ellipsoidal?
- How does the estimate change as N increases (20 $\rightarrow$ 200 $\rightarrow$ 2000)?
- Based on these plots, does the bivariate normal assumption seem reasonable?

Code

```
> library(MASS)

> fx.hat <- kde2d(x = x[,1], y = x[,2])

> fx.hat$x
> fx.hat$y
> fx.hat$z

> persp3d(x = fx.hat$x, y = fx.hat$y, z = fx.hat$z,
      col = "orange", xlab = "x1", ylab = "x2",
      zlab = "f(x)", xlim = c(10, 20),
      ylim = c(15, 25), zlim = c(0, 0.18))
```

The `kde2d()` function produces a "kernel density estimate" for the probability density based on the sampled data. You are not responsible for knowing how kernel density estimation works.
The actual normal distribution can be added to the plots as well:
Contour plots can be drawn as well; please see the corresponding program.

Standardized Data

Sometimes it is easier to work with data which are on the same scale. Standardized data can be used to convert the data to a unitless scale. Let

$$z_{rj} = \frac{x_{rj} - \hat{\mu}_j}{\sqrt{\hat{\sigma}_{jj}}}$$

for $r = 1, \ldots, N$ and $j = 1, \ldots, p$. The z_{rj} values together will have a sample mean of 0 and sample variance of 1 for each variable j.
We can put all of these values into a matrix:

$$Z = \begin{bmatrix} z_{11} & z_{12} & \cdots & z_{1p} \\ z_{21} & z_{22} & \cdots & z_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ z_{N1} & z_{N2} & \cdots & z_{Np} \end{bmatrix}.$$

Example: Stock Returns Data

For NVDA, AAPL, AMZN, and AMD returns, we standardize the variables:

```
> Z <- scale(stock[, 2:5])

> head(Z)

> cov(Z)

> cor(Z)

> colMeans(Z)
>
```

Questions

- Why is the estimated covariance matrix equal to the estimated correlation matrix?
- Notice the estimated correlation matrix is the same as the estimated correlation matrix for the original data. Why?

Example: Stock Returns Data (`stock_returns.csv`)

For NVDA, AAPL, AMZN, and AMD returns, we standardize the values:

```
> Z <- scale(stock[, 2:5])

> head(Z)
> NVDA       AAPL       AMZN        AMD
> 1   0.512384   0.218764   0.145332   0.620541
> 2  -0.845120  -0.312984  -0.420331  -0.910224
> 3   1.205438   0.782341   0.650112   1.432118
> 4   0.210335   0.145672   0.238994   0.498231
> 5  -1.120442  -0.745331  -0.932114  -1.365772
> 6   0.345228   0.198432   0.301552   0.702441

> cov(Z)
> NVDA  AAPL  AMZN   AMD
> NVDA   1.000 0.420 0.365 0.455
> AAPL   0.420 1.000 0.510 0.480
> AMZN   0.365 0.510 1.000 0.440
> AMD    0.455 0.480 0.440 1.000

> cor(Z)
> NVDA  AAPL  AMZN   AMD
> NVDA   1.000 0.420 0.365 0.455
> AAPL   0.420 1.000 0.510 0.480
> AMZN   0.365 0.510 1.000 0.440
> AMD    0.455 0.480 0.440 1.000

> colMeans(Z)
> NVDA       AAPL       AMZN        AMD
> 0.000000  0.000000  0.000000  0.000000
>
```

Questions

- Why is the estimated covariance matrix equal to the estimated correlation matrix?
- Notice the estimated correlation matrix is the same as the estimated correlation matrix for the original data. Why?

Eigenvectors and Multivariate Normal Distributions

Let λ_1 be the largest eigenvalue from Σ, let λ_2 be the 2nd largest eigenvalue from Σ, ..., and let λ_p be the smallest eigenvalue from Σ.
With respect to a bivariate normal distribution, the spread and direction of contours on a contour plot are related to the direction of the eigenvectors.

- The first eigenvector (corresponding to λ_1) points in the direction of the major axis of the ellipse created by the contours. This is the direction of the largest variability.
- The second eigenvector (corresponding to λ_2) points in the direction of the minor axis of the ellipse created by the contours. This is the direction of the smallest variability.

Example: Bivariate normal distribution

Let

$$\mathbf{x} \sim N_2\left(\begin{bmatrix}12\\18\end{bmatrix}, \begin{bmatrix}1.5 & 0.4\\0.4 & 1.1\end{bmatrix}\right)$$

as before.
Plotted below are the $f(x) = 0.01$ and 0.001 contours and the eigenvectors with a length of 10.

```
> mu <- c(12, 18)

> sigma <- matrix(c(1.5, 0.4,

                0.4, 1.1), nrow = 2, byrow = TRUE)

> save.eig <- eigen(sigma)

> save.eig$values
> [1] 1.640 0.960

> save.eig$vectors
> [,1]       [,2]
> [1,]   0.812345  -0.583201
> [2,]   0.583201   0.812345
>
```

The eigenvectors point in the direction of the ellipses' major and minor axes. The first eigenvector points in the direction of the largest variability. What if Var(x_2) is smaller? Let

$$\mathbf{x} \sim N_2\left(\begin{bmatrix}12\\18\end{bmatrix}, \begin{bmatrix}1.5 & 0.4\\0.4 & 0.6\end{bmatrix}\right)$$

and note that $|\Sigma| > 0$.
In fact, as λ_2 goes to 0, the contours will become a straight line. Thus, we could essentially just use one dimension to view resulting data from this distribution.

What if we had a trivariate normal distribution?

Plot 3×1 vectors in 3 dimensions with each axis corresponding to one of the three variable values. A 3D contour plot could be drawn as well, where a 3D ellipsoid would appear. The size of the 3D ellipsoid corresponds to the size of the eigenvalues: a small eigenvalue corresponds to a short axis, and a large eigenvalue corresponds to a long axis.

Pay special attention to examples where an eigenvalue is 0

What about the most general p-variate case?

Suppose there are 20 variables in a data set. A plot of these variables in 20 dimensions would be very difficult to do. If possible, we would prefer to view data values in 2 or 3 dimensions. Suppose that all of the estimated covariance matrix's eigenvalues were approximately 0 except for 2 or 3. This implies that the data can be viewed in 2 or 3 dimensions without losing much information.
This is essentially what principal components analysis allows us to do: view our data in a much smaller number of dimensions.

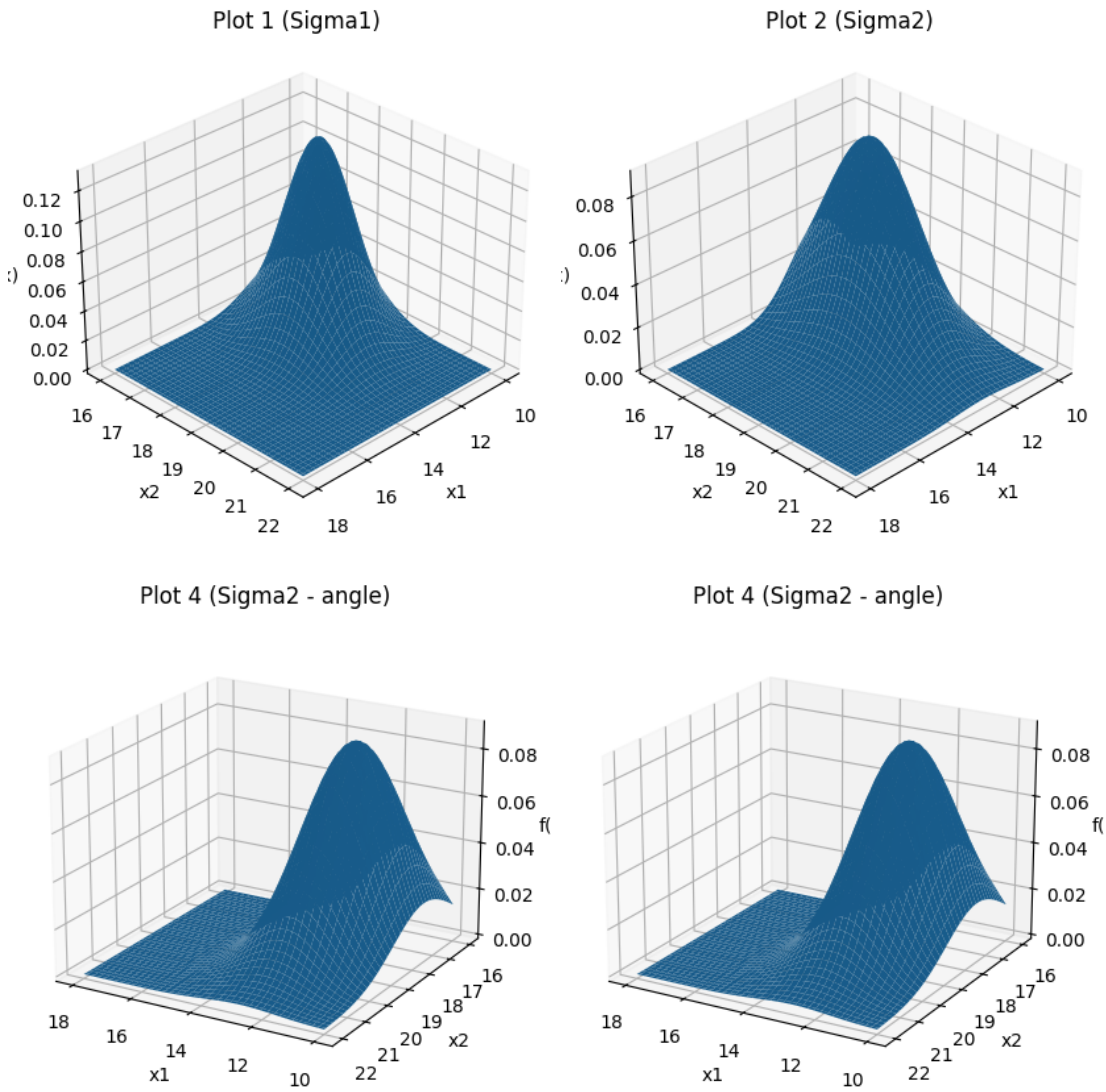

Figure 1: 3D surface plots of bivariate normal distributions under different covariance structures. Left column: $\Sigma_1 = \begin{bmatrix} 1.5 & 0.4 \\ 0.4 & 1.1 \end{bmatrix}$. Right column: $\Sigma_2 = \begin{bmatrix} 1.5 & 0.4 \\ 0.4 & 2.2 \end{bmatrix}$. Different viewing angles are shown for comparison.

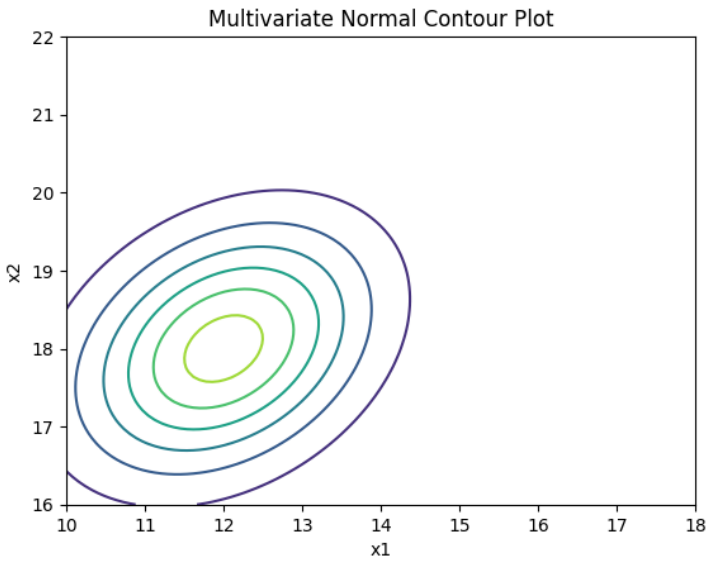

Figure 2: Multivariate normal contour plot

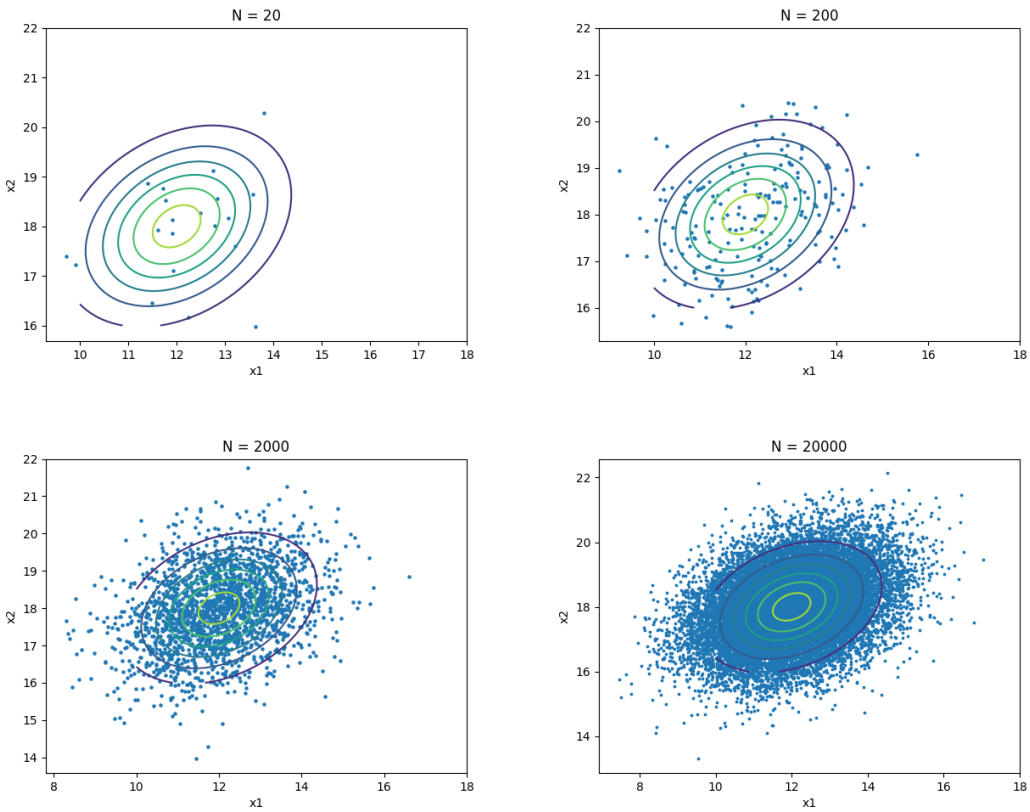

Figure 3: Multivariate normal contour plots with samples of size $N \in \{20, 200, 2000, 20000\}$.

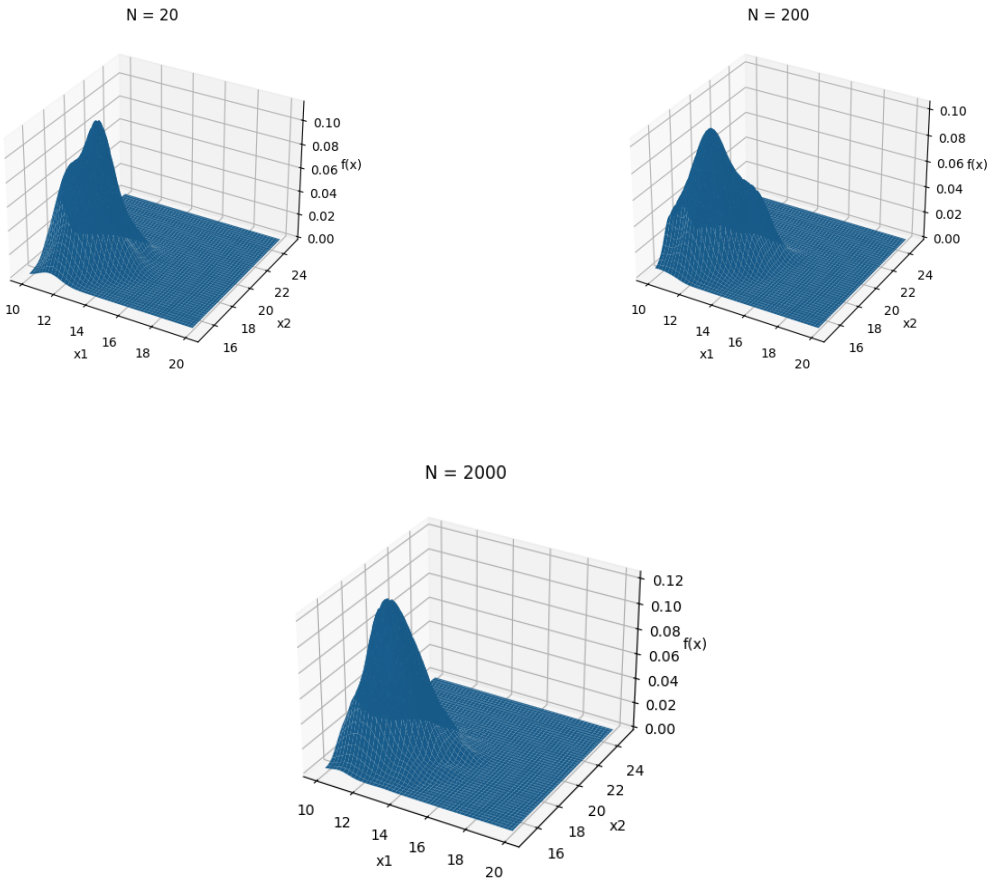

Figure 4: Kernel density estimates for different sample sizes ($N \in \{20, 200, 2000\}$).

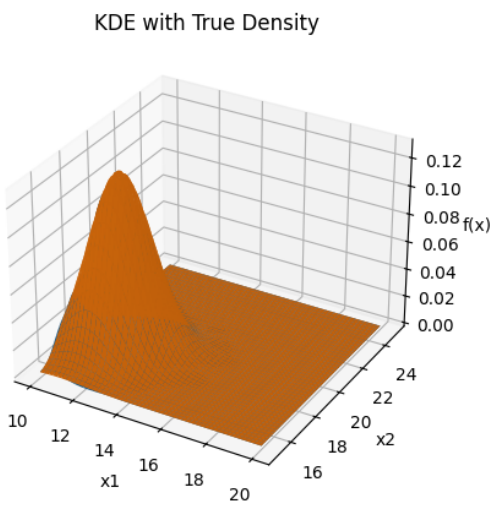

Figure 5: Kernel density estimate with true multivariate normal surface.

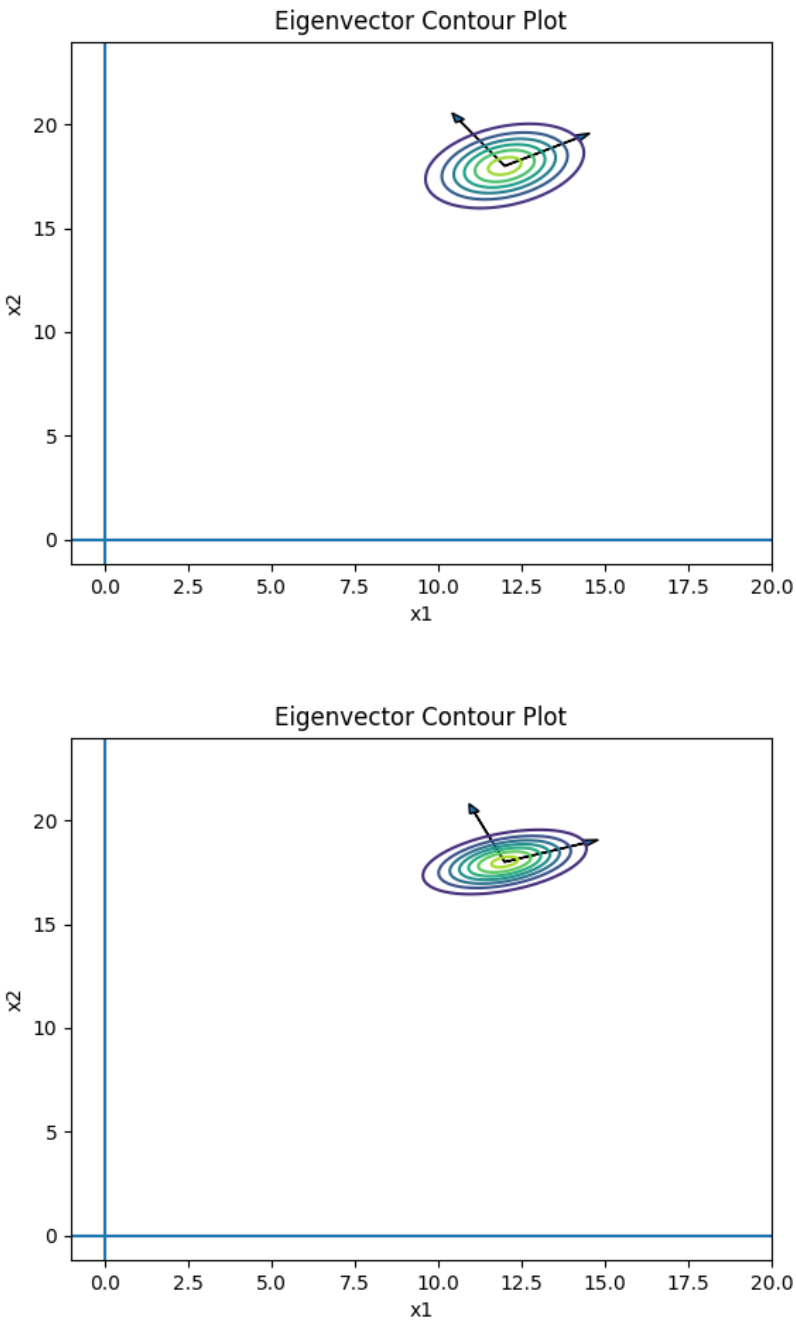

Figure 6: Contour plots with eigenvectors: original covariance (top) and reduced variance in x_2 (bottom).